

SOUTHEND, SK, Canada

WMO: 714510

Lat: 56.337N

Lon: 103.287W

Elev: 344

StdP: 97.26

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-38.7	-35.9	-42.5	0.1	-38.3	-39.7	0.1	-35.8	6.6	-17.6	6.3	-19.7	1.4	240	0.657

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.1	27.5	17.1	25.6	16.2	23.8	15.6	18.8	24.2	17.8	22.9	16.8	21.5	2.6	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.1	12.7	20.4	16.0	11.8	19.4	14.9	11.0	18.4	55.0	24.4	51.5	23.0	48.5	21.5	24.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.4	5.9	5.2	DB	-41.7	31.7	2.5	1.7	-43.5	32.9	-45.0	33.9	-46.4	34.9	-48.2	36.2	(4)
(5)			WB	-42.0	20.8	2.5	1.4	-43.8	21.8	-45.3	22.6	-46.7	23.4	-48.5	24.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.6	-21.1	-18.7	-11.4	-1.6	6.1	13.2	16.3	14.9	9.2	1.4	-9.8	-18.4	(6)	
	DBStd	14.69	8.46	7.91	8.40	7.03	5.14	4.02	3.07	3.38	4.36	4.46	7.06	8.60	(7)	
	HDD10.0	4751	964	803	664	349	143	14	1	3	67	269	593	880	(8)	
	HDD18.3	7294	1223	1037	922	598	379	160	77	116	276	526	843	1138	(9)	
	CDD10.0	528	0	0	0	2	22	110	197	154	42	1	0	0	(10)	
	CDD18.3	31	0	0	0	0	1	6	15	8	1	0	0	0	(11)	
	CDH23.3	491	0	0	0	1	26	105	217	124	18	0	0	0	(12)	
	CDH26.7	94	0	0	0	0	4	18	43	27	3	0	0	0	(13)	
	Wind	WSAvg	2.1	2.0	2.1	2.2	2.3	2.4	2.1	1.9	2.0	2.1	2.2	2.0	1.9	(14)
(15) Precipitation	PrecAvg	555	34	19	28	22	46	69	100	76	60	41	29	23	(15)	
	PrecMax	880	184	42	108	48	96	150	216	195	118	86	69	94	(16)	
	PrecMin	374	3	3	1	4	3	16	16	9	12	7	10	7	(17)	
	PrecStd	130	38	11	25	13	24	34	52	42	28	20	14	21	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.7	2.4	10.7	19.8	26.2	28.8	30.5	29.8	25.8	17.2	5.7	0.5	(19)	
		MCWB	-0.2	0.5	4.6	10.3	13.9	17.2	18.8	18.9	16.1	10.3	2.9	-0.1	(20)	
	2%	DB	-2.5	-0.3	6.9	15.2	22.7	26.1	27.5	26.7	22.0	12.9	2.7	-2.1	(21)	
		MCWB	-3.0	-1.2	2.6	7.6	11.9	15.6	17.5	17.4	14.6	8.4	1.2	-2.6	(22)	
	5%	DB	-5.4	-3.1	3.6	11.7	19.6	24.0	25.7	24.3	19.3	10.0	0.9	-4.5	(23)	
		MCWB	-5.7	-3.9	0.8	5.4	10.6	14.7	16.6	16.5	13.4	7.3	0.1	-4.9	(24)	
	10%	DB	-8.7	-6.2	0.9	8.7	16.3	21.8	23.9	22.1	16.6	7.8	-0.6	-6.4	(25)	
		MCWB	-9.0	-6.7	-0.7	3.9	9.2	13.9	16.3	15.8	12.5	6.0	-1.3	-6.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.2	0.7	5.2	10.6	14.3	20.0	20.5	19.8	17.5	11.6	3.2	-0.2	(27)	
		MCDB	0.7	1.6	10.1	19.2	23.4	25.5	26.1	25.5	23.9	14.7	5.1	0.2	(28)	
	2%	WB	-2.9	-1.4	3.0	7.6	12.6	17.4	19.0	18.4	15.5	9.1	1.4	-2.6	(29)	
		MCDB	-2.5	-0.7	6.4	14.2	19.9	22.1	24.4	23.8	20.4	12.0	2.4	-2.2	(30)	
	5%	WB	-5.6	-4.2	1.1	5.7	11.1	15.9	18.0	17.4	14.2	7.6	0.1	-4.9	(31)	
		MCDB	-5.3	-3.5	3.3	10.7	18.3	21.4	23.4	22.4	18.1	9.8	0.9	-4.5	(32)	
	10%	WB	-8.8	-7.0	-0.8	4.1	9.6	14.7	17.1	16.4	12.8	6.0	-1.2	-6.8	(33)	
		MCDB	-8.5	-6.4	1.0	8.2	15.2	20.1	21.8	20.5	16.2	7.8	-0.6	-6.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.8	12.3	14.2	13.2	12.7	12.4	12.1	11.2	10.2	7.1	7.7	8.7	(35)	
		MCDBR	11.4	12.7	15.4	16.8	19.4	17.0	16.2	15.4	14.9	11.8	7.8	9.2	(36)	
		MCWBR	10.8	11.4	11.4	9.8	10.0	8.0	7.5	7.6	8.4	7.9	6.5	8.7	(37)	
		MCDBR	11.1	12.3	13.8	15.2	16.8	13.6	13.2	13.2	13.2	11.0	7.4	8.9	(38)	
	5% WB	MCWBR	10.7	11.5	10.2	9.2	9.1	7.0	7.0	7.3	7.9	7.7	6.3	8.7	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.228	0.247	0.268	0.315	0.327	0.366	0.394	0.356	0.312	0.287	0.237	0.214	(40)	
	taud		2.255	2.301	2.305	2.352	2.493	2.413	2.356	2.477	2.585	2.563	2.386	2.204	(41)	
	Ebn at Noon		710	842	907	906	915	876	840	856	857	795	705	625	(42)	
	Edh at Noon		49	72	93	105	98	108	112	93	72	57	43	39	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.75	1.83	3.48	5.07	5.61	5.67	5.35	4.35	2.90	1.46	0.68	0.45	(44)	
	RadStd		0.08	0.12	0.22	0.34	0.56	0.46	0.36	0.34	0.37	0.19	0.07	0.06	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page